

**THE SEVENTH WILLIAM LOWELL PUTNAM
MATHEMATICAL COMPETITION**

May 24, 1947

Morning Session

1. If $\{a_n\}$ is a sequence of numbers such that for $n \geq 1$

$$(2 - a_n)a_{n+1} = 1,$$

prove that $\lim a_n$, as $n \rightarrow \infty$, exists and is equal to one. (page 223)

2. A real valued continuous function satisfies for all real x and y the functional equation

$$f(\sqrt{x^2 + y^2}) = f(x)f(y).$$

Prove that

$$f(x) = [f(1)]^{x^2}. \quad (\text{page 225})$$

3. Given this figure (see p. 23) and any two points Q_1, Q_2 in the plane not lying on any of the segments $s_1, s_2, \dots, s_6$, show that there does not exist a polygonal line P joining Q_1 and Q_2 such that:

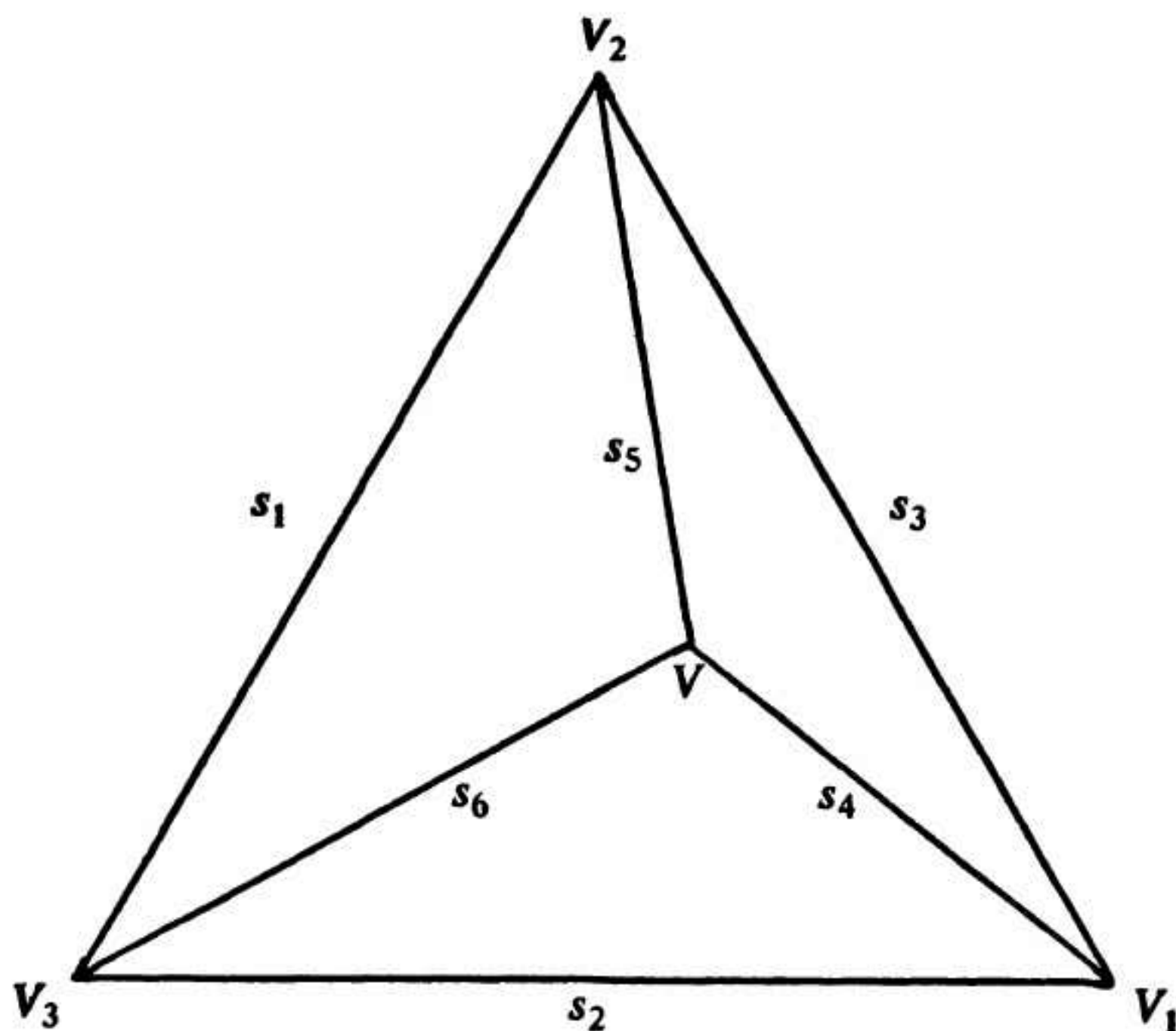
- (i) P crosses each $s_i, i = 1, 2, \dots, 6$, exactly once;
- (ii) P does not intersect itself;
- (iii) P does not pass through any vertex V_1, V_2, V_3, V_4 . (page 227)

4. A coast artillery gun can fire at any angle of elevation between 0° and 90° in a fixed vertical plane. If air resistance is neglected and the muzzle velocity is constant ($= v_0$), determine the set H of points in the plane and above the horizontal which can be hit. (page 229)

5. a_1, b_1, c_1 are positive numbers whose sum is 1, and for $n = 1, 2, \dots$ we define

$$a_{n+1} = a_n^2 + 2b_nc_n, \quad b_{n+1} = b_n^2 + 2c_na_n, \quad c_{n+1} = c_n^2 + 2a_nb_n.$$

Show that a_n, b_n, c_n approach limits as $n \rightarrow \infty$ and find these limits. (page 230)



Note. There is an unimportant error in the problem as it was presented to the contestants and as shown above: The “V” of the figure corresponds to the “ V_4 ” of the text.

6. A three-by-three matrix has determinant zero, and has the further property that the cofactor of any element is equal to the square of that element. (The cofactor of a_{ij} is $(-1)^{i+j}$ multiplied by the determinant obtained by striking out the i th row and j th column.) Show that every element in the matrix is zero. (page 234)

Afternoon Session

7. Let $f(x)$ be a function such that $f(1) = 1$ and for $x \geq 1$

$$f'(x) = \frac{1}{x^2 + f^2(x)}.$$

Prove that

$$\lim_{x \rightarrow \infty} f(x)$$

exists and is less than $1 + \pi/4$.

(page 235)

8. Let $f(x)$ be a differentiable function defined in the closed interval $(0, 1)$ and such that

$$|f'(x)| \leq M, \quad 0 < x < 1.$$

Prove that

$$\left| \int_0^1 f(x) \, dx - \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right) \right| \leq \frac{M}{n}. \quad (\text{page 236})$$

9. Let x, y be Cartesian coordinates in the plane. I denotes the line segment $1 \leq x \leq 3, y = 1$. For every point P on I , let P^* denote that point that lies on the segment joining the origin to P and such that the distance PP^* is equal to $1/100$. As P describes I , the corresponding point P^* describes a certain curve C^* . Let $l(I), l(C^*)$ be the lengths of I and C^* respectively. Which one of $l(I), l(C^*)$ is greater? Prove your answer. (page 238)

10. Given $P(z) = z^2 + az + b$, a quadratic polynomial of the complex variable z with complex coefficients a, b . Suppose that $|P(z)| = 1$ for every z such that $|z| = 1$. Prove that $a = b = 0$. (page 238)

11. a, b, c, d are distinct integers such that

$$(x - a)(x - b)(x - c)(x - d) - 4 = 0$$

has an integral root r . Show that $4r = a + b + c + d$. (page 240)

12. C is a fixed point on OZ and U, V are variable points on OX, OY respectively, where OX, OY, OZ are mutually orthogonal lines. Find the locus of a point P such that PU, PV, PC are mutually orthogonal. (page 241)